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The Algebra of Narratives and the Monitoring of Their Relative Dominance

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Abstract

We propose an algebraic methodology for measuring the relative prevalence of competing narratives within a time-evolving corpus of texts. Standard cosine similarity between article embeddings and narrative vectors fails to isolate the idiosyncratic contribution of each narrative, because narratives sharing a common topic produce strongly correlated vector representations. We address this by orthogonalizing the narrative vectors via a global max-min optimization on the Stiefel manifold, which neutralizes shared semantic components while preserving each narrative's specific direction. We ground this geometric procedure in a linear generative model for article embeddings that provides an unbiased estimator of narrative prevalence and formally explains why the naive approach suffers from a correlation bias. Empirical validation on the publicly available News Category Dataset (Huff Post, 2012–2018) confirms that the orthogonalized estimator recovers narrative-specific dynamics aligned with known historical events, whereas the naive cosine projection yields synchronised, uninformative time series. **[TODO: Refine abstract once results section is complete.]**

Keywords: narrative analysis, Stiefel manifold, text embeddings, orthogonalization, semantic similarity

[CAL: juste pour vérifier que tout est coherent]

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1. Positioning

Since the emergence of vector representations of texts, their power has been evidenced by the structure of distance in the Euclidean space to which they belong. Early demonstrations include the native clustering of Self-Organizing Map (SOM) prototypes on Grimm tales (Honkela, Pulkki, and Kohonen 1995) and the celebrated *king* − *man* + *woman* = *queen* relation satisfied by doc2vec embeddings (Mikolov, Yih, and Zweig 2013). Modern bidirectional encoders such as BERT have turned semantic similarity into a practical primitive, notably in Retrieval-Augmented Generation (RAG) pipelines (Wang et al. 2024). Concurrently, the concept of *narrative competition* has gained traction in economic literature (Shiller 2020): narratives are the mean field through which opinions form, and agents condition their decisions on the perceived relative dominance of narratives. Institutional actors exploit this dynamic; for instance, the Bank of England instruments narratives to guide market expectations (Naef 2024). Market participants continuously probe and reshape narratives in a closed feedback loop that renders straightforward causal analysis difficult.

The narrative viewpoint differs fundamentally from sentiment analysis. Rather than classifying a document as positive or negative—an approach that conflates semantic and frequentist synonyms (Yang et al. 2026)—it uses the alignment of a text with the vector representation of a narrative as a measure of *importance*. Formally, the cosine similarity between a news article and a narrative takes values close to +1 when they are perfectly aligned, close to −1 when contradictory, and close to zero when semantically independent in the high-dimensional embedding space.

This paper addresses the following question: how can one monitor, over time and at scale, the relative prevalence of competing narratives within a large corpus of text? Existing approaches largely fall into three families. The first relies on keyword or dictionary-based counting: canonical examples include the Economic Policy Uncertainty index of (Baker, Bloom, and Davis 2016), built from keyword counts in newspaper archives, and the finance-specific sentiment dictionaries of (Loughran and McDonald 2011). Being built on exact vocabulary matches, this family offers limited semantic granularity: a narrative expressed through paraphrase or implication, without the target keywords, goes undetected. The second relies on unsupervised topic modeling, typically via Latent Dirichlet Allocation applied to financial or economic news (Bybee et al. 2020; Larsen and Thorsrud 2019), or, closer to our setting, on text-semantics pipelines that extract narrative statements from syntactic structure (Ash, Gauthier, and Widmer 2021). Although richer than raw keyword counts, this family remains fundamentally a bag-of-words approach: topics are inferred from word

co-occurrence statistics across the corpus, with no representation of synonymy or paraphrase and no contextualization of polysemous words, and more importantly for our purposes, a narrative of interest cannot be specified *a priori* by a short description; it must instead be statistically prevalent enough in the corpus to emerge as a topic on its own. The third relies on supervised classifiers trained to detect a predefined narrative from labeled examples: for instance, (Coan et al. 2021) build a two-level taxonomy of climate-change contrarian claims and train a classifier on tens of thousands of manually annotated paragraphs; + papier de Salomé Do et Étienne Ollion aumgented scientist **[JAMM: j'avais lu ce papier qui décrit exactement une approche par classification avec BERT, est-ce pertinent ?]**. [CAL: surtout si on ajoute l'exemple de l'IRan mais ça peut faire beaucoup à ajouter, je me propose de faire un autre papier basé sur les classifications pour étendre celui-ci, la référence sera la bienvenues] Unlike the first two families, such classifiers, built on contextual embeddings, can achieve fine-grained semantic resolution but only at the cost of substantial human labeling effort for every narrative of interest, and they do not generalize to a newly introduced narrative without retraining.

The qualitative leap we seek is the ability to measure narrative prevalence at a fine level of semantic nuance without either family's cost: neither reduced to a fixed vocabulary or a corpus-driven topic, nor requiring labeled examples per narrative. Just as importantly, we seek to compare the resulting measurements *across* narratives. This requires prevalence estimates for different narratives to sit on a common, mutually comparable scale, not merely to be individually well-behaved in time. The methodology we propose specifies a narrative *a posteriori* by a short descriptive text, and measures its prevalence directly against an unlabeled corpus, in the spirit of zero-shot, embedding-based text classification (Schopf, Braun, and Matthes 2023).

Challenges. Measuring relative narrative strength poses two distinct challenges.

1. **Idiosyncratic vs. common co-movements.** Even apparently opposed narratives share a common semantic direction (the *elections* axis) that causes their cosine scores to rise and fall in unison for reasons unrelated to their proper meaning. Disentangling idiosyncratic from common variation is therefore essential.
2. **Comparability across narratives.** Comparing the relative prevalence of two narratives requires their estimates to be measured on a common scale. With raw, non-orthogonal candidate vectors, this comparability fails on two counts: the measured signal for one narrative is contaminated by leakage from every other narrative sharing semantic content with it, and nothing ties the scale on which one narrative's presence is measured to that of another, since each is built from an independently chosen descriptive text.

Contributions. We propose an algebraic framework that simultaneously addresses both challenges. Our key contributions are:

- A linear generative model for article embeddings, articulated around the abstract, idealized concept of a *pure narrative*, a collection of orthonormal vectors that exists only in the latent embedding space, not as any literal piece of text. The model provides a rigorous statistical foundation for measuring narrative prevalence, yielding an unbiased estimator under pure narratives and formally exposing the correlation bias incurred when monitoring narrative prevalence via raw cosine similarity to non-orthogonal candidate texts (Section 2).
- Building on this diagnosis, a global orthogonalization procedure on the Stiefel manifold that constructs, from an arbitrary collection of candidate narrative texts, the corresponding collection of pure narratives: for each candidate, a representative vector that is close to the original text while being orthogonal to all others (Section 4).

- An empirical validation on a publicly available labeled news dataset, including a counterfactual study contrasting the naive and orthogonalized measures, demonstrating qualitative and quantitative agreement with known historical events (Section 5).

It is not the first time algebra has been used in the space of embedding to remove biases or common effects. This has been notability done to remove gender biases of language models (Dev and Phillips 2019; Dawkins et al. 2024). The main difference is that whereas such operations on gender bias has to pre-defined polarities to orthogonalize too (the vector of the "he–she difference" or a set of a gender-paired sentences), in the case of a collation of narratives the directions to orthogonalize to are only implicitly defined. That is why we have to turn to the Stiefel manifold: looking for representative of narratives that are as close as the original as possible but as orthogonal to the rest of others as possible.

2. A Generative Model for Narrative Embeddings

2.1 Pure Narratives: an Idealized Abstraction

We take a *narrative* to be an abstract idea, claim, or framing about the world: an assertion, an interpretation of events, or a story that can be told about them. As a result, a piece of text can be a *representation* of a given narrative without being the narrative itself. Text is not the only possible representation, however. Language models make it possible to build a bridge between the qualitative world of sentences and words and the quantitative world. A narrative can then be represented as a vector in a latent space $\mathcal{H} \subset \mathbb{R}^d$, so that comparing or measuring narratives reduces to standard geometric operations on $\mathcal{H}$. This is a truly Platonist claim: just as Plato's Ideas live in their own world and language is only a support for them, *narratives* live in their own world, and texts or vectors are only supports for them.

However, by abuse of terminology, we will sometimes identify a narrative with its representation (either a text or a vector).

[CAL: illustration faut peut être choisir une typo pour les narratifs, si on met tout en italique ça peut être compliqué; il faut choisir entre (apr exemple): "*Welfare State Under Pressure*" vs "Welfare State Under Pressure" vs Welfare State Under Pressure vs Welfare State Under Pressure (j'aime bien le dernier); j'ai fait une macro narr]

For instance, the left panel of Table 1 shows the cosine similarities (for modernBERT embeddings) of the following three English representations of the narrative Extreme Heat and Emergency, Policy Response:

1. Record-breaking temperatures across southern Europe this summer triggered unprecedented heatwaves, forcing governments to activate emergency climate plans and accelerate the transition toward renewable energy sources in order to meaningfully reduce carbon emissions and limit further global warming.
2. Record-breaking temperatures across southern Europe this summer fueled unprecedented heatwaves, prompting governments to implement emergency climate measures and fast-track the adoption of renewable energy to curb carbon emissions and mitigate further global warming.
3. Southern Europe faced extreme heat this summer, with record-breaking temperatures triggering a climate emergency and driving governments to intensify policy responses, accelerate the shift to clean energy, and implement measures to reduce carbon emissions and limit future global warming..

Their pairwise similarities are high, it explains why vectors of embedding, despite they live in a high dimension (768 for modernBERT), can be used as algebraic representatives of the narrative itself.

Nevertheless, if we take two other arbitratry narratives of Appendix 1:

4. NATO Eastern Flank and the Russia–Ukraine War: The ongoing military conflict between Russia and Ukraine continues to destabilize the European geopolitical order, compelling NATO allies to reinforce their eastern flank with additional troops and to significantly increase national defense budgets in response to sustained and escalating Russian aggression.
5. IMF Forecasts and Global Growth Slowdown: Global economic growth is decelerating at a concerning pace according to the latest International Monetary Fund projections, which identify trade tensions, rising public debt levels, and slowing emerging market performance as the primary drivers behind deteriorating worldwide economic prospects.

The right panel of the same table show there pairwise cosine similarity is lower (around 0.92 in place of 0.96), but keeps very high. It is due to common English grammar, sentence building, etc. Structural semantic similarities are hiding the underlying differences between narratives.

This observation of apparent similarity justifies the following definition.

Table 1. Pairwise Cosine Similarity Matrix (ModernBERT) for the narratives of the example inspired from Appendix 1.

	Narrative 1	Narrative 2	Narrative 3		Narrative 1	Narrative 4	Narrative 5
Nar. 1	1.00	0.95	0.96	Nar. 1	1.00	0.92	0.93
Nar. 2	0.95	1.00	0.99	Nar. 4	0.92	1.00	0.91
Nar. 3	0.96	0.99	1.00	Nar. 5	0.93	0.91	1.00

Collection of Pure Narratives Rather than studying a single narrative in isolation, this paper is concerned with studying relative prevalence among several narratives. We accordingly introduce the notion of a *collection of pure narratives*, whose representations live in the embedding space $\mathcal{H}$. A collection of pure narratives is a collection of narratives that are semantically independent from one another: each retains only an idiosyncratic component, separate from whatever is common across the collection. Purity in this sense is not a property of a single narrative taken in isolation, it is only meaningful relative to the collection it belongs to. In the embeddings space, this translate as followed:

Definition 1 (Collection of pure narratives)**.** *A collection $V_1, \dots, V_K$ of narratives in $\mathcal{H} \subset \mathbb{R}^d$ is a collection of* pure narratives *if the collection $(V_i)_{1 \leq i \leq K}$ is orthonormal.*

Pure narratives should be understood as an idealized abstraction rather than a text we expect to observe or write down directly: we do not claim that any collection of texts is an exact representation of a collection of pure narratives, and we make no attempt to produce one. What matters at this stage is only that such an orthonormal collection *exists* in $\mathcal{H}$; Section 4 then shows how to recover, from any (impure) list of narrative-representing texts, a collection of vectors that is a good practical proxy for it.

In principle a collection of pure narratives for our previous example of (1) Extreme Heat and Emergency, PolicyResponse, (4) NATO Eastern Flank and the Russia–Ukraine War, and (5) IMF Forecasts and Global Growth Slowdown should have a right panel of Table 1 that is close to a diagonal of ones: they should be pairwise orthogonal (in the cosine similarity sense).

2.2 A Generative Model on the Embedding Space

We start by explaining and defining clearly the notion of *prevalence* for pure narratives; in Section 3.1 we show how to orthogonalize a family of non-pure narratives to use the same notions on any family of representatives of narratives.

Given a family of pure narratives of interest $(V_i)_{1\leq i\leq K}$, our goal is to study their relative prevalence within a corpus of texts. We introduce a probabilistic generative model for this purpose.

Setup. We fix, for each $i \leq K$, a function $p_i : [0,\infty) \to [0,1]$. We consider the random variables $(\alpha_1, ..., \alpha_K, \epsilon)$. We place ourselves on the probability space $(\Omega, \mathcal{F}, \mathbb{P})$ generated by the law of $(\alpha_1, \ldots, \alpha_K, \varepsilon)$ such that: to an occurrence $\omega \in \Omega$ that occurs at time $\tau(\omega)$, we associate $\alpha_i(\omega) \in \{0,1\}$ $(i \leq K)$ and $\varepsilon(\omega) \in \mathcal{V}^\perp := \mathrm{Vect}(V_1, \ldots, V_K)^\perp$, whose joint law, conditional on $\tau(\omega) = t$, is: the $\alpha_i(\omega)$ are Bernoulli with parameter $p_i(t)$, and $\varepsilon(\omega) \sim \mathcal{N}(0, \sigma_\varepsilon^2)$ on $\mathcal{V}^\perp$, independent of $(\alpha_i(\omega))_{i\leq K}$.

Adding time via $\tau(\omega)$ equips $(\Omega, \mathcal{F}, \mathbb{P})$ with a filtration $(\mathcal{F}_t)_{t\geq 0}$: given a corpus of occurrences $\omega_1, \omega_2, \ldots$, $\mathcal{F}_t$ records which occurrences have arisen by time t, together with the mark $(\alpha_i(\omega_n))_{i\leq K}, \varepsilon(\omega_n)$ of every occurrence n with $\tau(\omega_n) < t$. By construction, τ is an $(\mathcal{F}_t)$-stopping time.

We introduce the random variable X and assume that $X(\omega)$ decomposes as

$$X(\omega) = \sum_{i=1}^{K} \alpha_i(\omega)\, V_i + \varepsilon(\omega). \tag{1}$$

To keep the notation light, we drop the dependence on ω: given a database of N occurrences, [CAL: c'est là qu'il faut introduire une mesure (on va prendre l'esperence suivant cette mesure) et un échantillon $(X_n)_n$ tiré de façon iid suivant cette loi (voir plus loin)] we write $X_n := X(\omega_n)$, $\alpha_{i,n} := \alpha_i(\omega_n)$, $\varepsilon_n := \varepsilon(\omega_n)$, and $t_n := \tau(\omega_n)$, so that (1) reads

$$X_n = \sum_{i=1}^{K} \alpha_{i,n}\, V_i + \varepsilon_n, \qquad n = 1, \ldots, N. \tag{2}$$

Since $(V_i)_{i\leq K}$ are orthonormal, they span a subspace $\mathcal{V} := \mathrm{Vect}(V_1, \ldots, V_K)$, and $\mathcal{H} = \mathcal{V} \oplus \mathcal{V}^\perp$. Consequently, every $X_n \in \mathcal{H}$ admits a unique decomposition $X_n = X_n^{\|} + X_n^{\perp}$ with $X_n^{\|} \in \mathcal{V}$ and $X_n^{\perp} \in \mathcal{V}^\perp$, matching (2) with $\alpha_{i,n} = V_i^\top X_n$ and $\varepsilon_n = X_n^{\perp}$.

Qualitative interpretation. If one considers a corpus of texts (for instance a corpus of News) with N occurrences, the collection $(X_n)_{1\leq n\leq N}$ represents the corpus, more precisely, each X_n is the vector embedding of a text (for instance of an article). The variable $\alpha_{i,n}$ indicates whether narrative V_i is present in text n. The residual ε_n captures everything in the text unrelated to the K narratives of interest: for instance, author style, secondary facts, or unmodeled topics. As it lies in $\mathcal{V}^\perp$, it carries no component along any narrative of interest.

2.3 Temporal Conditioning and Prevalence

Time discretization Time is discretized into periods indexed by $t \in \{1, \ldots, T\}$, corresponding to the values taken by the stopping time τ once the observation window is partitioned into days, weeks, or months, without fixing *a priori* the granularity. For each period t, we denote by

$$\mathcal{C}_t := \{n : \tau(\omega_n) = t\}, \qquad n_t := |\mathcal{C}_t| \tag{3}$$

the set and number of articles published during period t. Since τ is an $(\mathcal{F}_t)$-stopping time, $\{\tau(\omega_n) = t\}$ is $\mathcal{F}_t$-measurable for each n, and hence so are $\mathcal{C}_t$ and n_t.

Conditional prevalence. Recall from Section 2.2 that $p_i(t)$ denotes the Bernoulli parameter governing the presence of narrative i at time t: $\mathbb{E}[\alpha_{i,n} \mid \mathcal{F}_t] = p_i(t)$ for any $n \in \mathcal{C}_t$. This quantity is not directly observable, since $\alpha_{i,n}$ is itself latent. Only the embedding X_n is.

But, we leverage the properties of our model: the orthonormality of the V_k and the fact that $\varepsilon_n \in \mathrm{Vect}(V_1, ..., V_K)^\perp$ to obtain:

$$\mathbb{E}\big[V_i^\top X_n \mid \mathcal{F}_t\big] = \sum_{k=1}^{K} \underbrace{\mathbb{E}\big[\alpha_{k,n} \mid \mathcal{F}_t\big]}_{=p_k(t)} \underbrace{V_i^\top V_k}_{=\delta_{ik}} + \mathbb{E}\big[\underbrace{V_i^\top \varepsilon_n}_{=0} \mid \mathcal{F}_t\big] = p_i(t), \tag{4}$$

[CAL: je ne suis pas hyper fan de $\mathbb{E}\big[V_i^\top X_n \mid \mathcal{F}_t\big]$ (l'espérence est sur quoi?) **il faut écrire $\mathbb{E}\big[V_i^\top X \mid \mathcal{F}_t\big]$, où $X \in$ notre corpus avec la distribution naturelle, le conditionnement fait la sélection des points (car iid)**]

This shows that $p_i(t)$ coincides with the conditional expectation of the observable projection $V_i^\top X_n$, which is what makes it estimable in practice.

Proposition 1.

$$\mathbb{E}\big[V_i^\top X_n \mid \mathcal{F}_t\big] = p_i(t). \tag{5}$$

Definition 2 (Prevalence of narratives in texts)**.** *The previous proposition gives a constructive definition of* the prevalence of narrative V_i in text X*, that is*

$$\hat{\alpha}_i := \mathbb{E}\big[V_i^\top X\big]. \tag{6}$$

This is a well defined notion and can be conditioned by a filtration to get *the prevalence of narrative V_i in a corpus of text at t*, that is given by equality (5). Note that thanks to our model, the prevalence, that can be read as the contribution of the pure narrative V_i to the text X, is nothing more than a scalar product, i.e. a cosine similarity. We show in Section 3.1 how to extend this estimate to families of non-pure narratives.

Absolute frequency. The absolute frequency of narrative i at time t jointly captures narrative intensity and publication volume:

$$q_i(t) := \mathbb{E}\big[V_i^\top X_n \cdot \mathbf{1}_{\tau(\omega_n)=t}\big]. \tag{7}$$

Co-occurrence. For $i, j \leq K$, we define the co-occurrence of narratives i and j at time t as

$$p_{ij}(t) := \mathbb{E}\big[\alpha_{i,n}\alpha_{j,n} \mid \mathcal{F}_t\big], \qquad n \in \mathcal{C}_t, \tag{8}$$

the conditional probability that an article published at t carries both narratives i and j simultaneously; note $p_{ii}(t) = p_i(t)$. As with prevalence, $p_{ij}(t)$ is not directly observable, since $\alpha_{i,n}, \alpha_{j,n}$ are themselves latent. The following proposition shows it is recovered exactly from the observable product of projections $(V_i^\top X_n)(V_j^\top X_n)$.

Proposition 2. *For any $n \in \mathcal{C}_t$,*

$$(V_i^\top X_n)(V_j^\top X_n) = \alpha_{i,n}\alpha_{j,n} \quad \textit{a.s.}, \qquad \textit{and hence} \qquad \mathbb{E}\big[(V_i^\top X_n)(V_j^\top X_n) \mid \mathcal{F}_t\big] = p_{ij}(t). \tag{9}$$

Proof. Since $V_i, V_j \in \mathcal{V}$ and $\varepsilon_n \in \mathcal{V}^\perp$ identically, $V_i^\top \varepsilon_n = V_j^\top \varepsilon_n = 0$ almost surely. Hence, using (2) and $V_k^\top V_l = \delta_{kl}$, $V_i^\top X_n = \alpha_{i,n}$ and $V_j^\top X_n = \alpha_{j,n}$ exactly. The product identity follows immediately, and taking $\mathbb{E}[\cdot \mid \mathcal{F}_t]$ recovers $p_{ij}(t)$ by definition. □

2.4 Empirical Estimators

This naturally leads to unbiased empirical estimators for the prevalence, the absolute frequency, and the co-occurrence.

$$\hat{p}_i(t) = \frac{1}{n_t} \sum_{n \in \mathcal{C}_t} V_i^\top X_n, \tag{10}$$

$$\hat{q}_i(t) = \frac{1}{N} \sum_{n \in \mathcal{C}_t} V_i^\top X_n = \frac{n_t}{N} \hat{p}_i(t). \tag{11}$$

$$\hat{p}_{ij}(t) = \frac{1}{n_t} \sum_{n \in \mathcal{C}_t} (V_i^\top X_n)(V_j^\top X_n). \tag{12}$$

[JAMM: Peut-être qu'on pourra supprimer ces preuves qui sont quasi immédiates][CAL: ou peut juste les garder en appendix au cas où]

Unbiasedness of $\hat{p}_i(t)$. Conditioning on $\mathcal{F}_t$:

$$\mathbb{E}\left[\hat{p}_i(t) \mid \mathcal{F}_t\right] = \frac{1}{n_t} \sum_{n \in \mathcal{C}_t} \mathbb{E}\left[V_i^\top X_n \mid \mathcal{F}_t\right] = \frac{1}{n_t} \sum_{n \in \mathcal{C}_t} p_i(t) = p_i(t). \tag{13}$$

By the tower property, $\hat{p}_i(t)$ is therefore an unbiased estimator of $p_i(t)$:

$$\mathbb{E}\left[\hat{p}_i(t)\right] = \mathbb{E}\Big[\mathbb{E}\left[\hat{p}_i(t) \mid \mathcal{F}_t\right]\Big] = p_i(t). \tag{14}$$

Unbiasedness of $\hat{q}_i(t)$. Because N is fixed, we can exchange expectation and summation directly:

$$\mathbb{E}\left[\hat{q}_i(t)\right] = \frac{1}{N} \sum_{n=1}^{N} \mathbb{E}\left[V_i^\top X_n \cdot \mathbf{1}_{\tau(\omega_n)=t}\right] = q_i(t). \tag{15}$$

Unbiasedness of $\hat{p}_{ij}(t)$. By Proposition 2, $(V_i^\top X_n)(V_j^\top X_n) = \alpha_{i,n}\alpha_{j,n}$ almost surely for every $n \in \mathcal{C}_t$, so conditioning on $\mathcal{F}_t$:

$$\mathbb{E}\left[\hat{p}_{ij}(t) \mid \mathcal{F}_t\right] = \frac{1}{n_t} \sum_{n \in \mathcal{C}_t} \mathbb{E}\left[(V_i^\top X_n)(V_j^\top X_n) \mid \mathcal{F}_t\right] = \frac{1}{n_t} \sum_{n \in \mathcal{C}_t} p_{ij}(t) = p_{ij}(t), \tag{16}$$

and, by the tower property, $\hat{p}_{ij}(t)$ is an unbiased estimator of $p_{ij}(t)$:

$$\mathbb{E}\left[\hat{p}_{ij}(t)\right] = \mathbb{E}\Big[\mathbb{E}\left[\hat{p}_{ij}(t) \mid \mathcal{F}_t\right]\Big] = p_{ij}(t). \tag{17}$$

3. Failure of the Naive Approach: why orthonormality matters

In Section 4.5 we made the difference between *representatives of narratives*, that are sentences describing in human language the targeted narrative, and we are defined *collections of pure narratives*, namely set of orthogonal narratives. In Section 2.2 we have shown that is the straightforward to build unbiased estimates of the *prevalence of narratives in a corpus of texts*.

Nevertheless with have illustrated in the same Section 4.5 that representatives of different narratives (we took the example of (1) *Extreme heat*, (2) *Russia-Ukraine war* and (3) *IMF Forecasts*) are not at all

orthogonal: they have a lot in common due to the English structure and the common semantic to be used to describe clearly any narrative in one short paragraph.

This section show why not using a collection of pure narratives would fail at estimating the prevalence and other quantities of interest. Not having access to a pure version of the collection of narratives prevent to simply estimate the importance (or joined importance) of narratives of interest in texts. ~~At this stage, one could ask what's the point in having the orthonormality assumption, and having designed a generative model for text embeddings. Indeed, let's return to our baseline objective: monitoring the relative prevalence of a collection of narratives $(V_1, \dots, V_K)$ understood here in the abstract sense of Section~~ 2.2.

~~One could claim that a simple way to proceed would be to write down a textual representation of each V_l and pass it through a language model, obtaining candidate vectors $N_l \in \mathcal{H}$. Empirically, such candidates do not form an orthonormal family: normalizing them to unit norm is immediate, but nothing in how language models produce embeddings guarantees that they come out pairwise orthogonal.~~

~~In other words,~~ the theoretical framework developed in Section 2 lets us establish precisely why this approach fails. We quantify it both at the level of prevalence (a first-moment quantity, Section 3.1) and at the level of co-occurrence, Section 3.2).

3.1 Prevalence

[TODO: Même question: garde-t-on les preuves ici ou en annexe][CAL: je suis d'accord, à transferer en appendix]

Let $(N_i)_{1\le i\le K} \in \mathcal{H}$ be a family of unit-vector embeddings that represents our studied narratives. The family is not asssumed to be orthogonal. Analogously to Section 2.3, the ~~naive~~ natural approach measures the prevalence of N_i at time t as

$$\pi_{N_i}(t) := \mathbb{E}\big[N_i^\top X_n \mid \mathcal{F}_t\big], \qquad n \in \mathcal{C}_t. \tag{18}$$

[CAL: attention: même remarque que plus haut, **il faut remplacer X_n par X** car l'espérance est sur la mesure dans l'ensemble des textes]

Since N_i and X_n are both unit-norm, $N_i^\top X_n$ is exactly their cosine similarity; $\pi_{N_i}(t)$ is then the average cosine similarity between narrative i's candidate embedding and the articles published during period t, precisely the quantity the naive approach would compute in practice.

Since $(V_k)_{k\le K}$ is an orthonormal basis of $\mathcal{V}$, N_i decomposes uniquely as

$$N_i = \sum_{k=1}^{K} c_{ik}\, V_k + N_i^{\perp}, \qquad c_{ik} := N_i^\top V_k, \quad N_i^{\perp} \in \mathcal{V}^{\perp}. \tag{19}$$

Proposition 3 (Naive prevalence under an impure candidate)**.**

$$\pi_{N_i}(t) = c_{ii}\, p_i(t) + \sum_{k\neq i} \underbrace{c_{ik}\, p_k(t)}_{\textit{leakage}}. \tag{20}$$

Proof. Using (19) and $X_n = \sum_l \alpha_{l,n} V_l + \varepsilon_n$,

$$\begin{aligned} N_i^\top X_n &= \Big(\sum_k c_{ik} V_k + N_i^{\perp}\Big)^\top \Big(\sum_l \alpha_{l,n} V_l + \varepsilon_n\Big) \\ &= \sum_k c_{ik}\, \alpha_{k,n} + (N_i^{\perp})^\top \varepsilon_n, \end{aligned}$$

using $V_k^\top V_l = \delta_{kl}$, $N_i^{\perp} \perp V_l$, and $V_k \perp \varepsilon_n$. Taking $\mathbb{E}[\cdot \mid \mathcal{F}_t]$, the term $\mathbb{E}[(N_i^{\perp})^\top \varepsilon_n \mid \mathcal{F}_t] = 0$ since ε_n is independent of $\mathcal{F}_t$ and centered, and $\mathbb{E}[\alpha_{k,n} \mid \mathcal{F}_t] = p_k(t)$, giving the result. □

Interpretation. When $(N_i)_i$ is pure, $c_{ii} = 1$ and $c_{ik} = 0$ for $k \neq i$, and $\pi_{N_i}(t) = p_i(t)$ exactly, recovering the unbiased estimator of Proposition 1. For an impure candidate, however, $\pi_{N_i}(t)$ mixes the true prevalence $p_i(t)$ with every other narrative's prevalence $p_k(t)$, weighted by the leakage coefficient $c_{ik} = N_i^\top V_k$. If a historical event drives up the prevalence of an unrelated but semantically connected narrative k ($p_k(t)$ increases), the naive indicator $\pi_{N_i}(t)$ increases mechanically, creating the illusion that narrative i itself has become more prevalent.

3.2 Co-occurrence of Two Narratives

Recall from Section 2.3 that the true co-occurrence of narratives i and j at time t, $p_{ij}(t) := \mathbb{E}[\alpha_{i,n}\alpha_{j,n} \mid \mathcal{F}_t]$, coincides exactly with $\mathbb{E}[(V_i^\top X_n)(V_j^\top X_n) \mid \mathcal{F}_t]$ for pure narratives (Proposition 2). For arbitrary candidates $W_i, W_j \in \mathcal{H}$, the naive approach measures their co-occurrence by transplanting this same estimator to the candidates it actually has access to *i.e* language-model embeddings of a textual description of each narrative, rather than the (unobservable) pure narratives themselves. Concretely, for each article n, $X_n^\top W_i$ is the cosine similarity between the article and candidate i; their product is large only when the article aligns strongly with *both* candidates at once, and small otherwise, so that averaging this product over the pool of articles published at t is, intuitively, a natural proxy for how often (and how strongly) narratives i and j jointly manifest in the corpus:

$$M_{ij}(t) := \mathbb{E}\big[(X_n^\top W_i)(X_n^\top W_j) \mid \mathcal{F}_t\big], \qquad n \in \mathcal{C}_t. \tag{21}$$

Proposition 4 (Naive co-occurrence under impure candidates)**.** *Write* $W_i = \sum_k c_{ik} V_k + \eta_i$ *and* $W_j = \sum_l c_{jl} V_l + \eta_j$ *as in* (19)*. Then*

$$M_{ij}(t) = c_{ij}^2 p_{ij}(t) + \underbrace{\sum_{k\neq i,j}\sum_{l\neq i,j} c_{ik}\, c_{jl}\, p_{kl}(t)}_{\textit{signal leakage}} + \underbrace{\sigma_\varepsilon^2\, \eta_i^\top \eta_j}_{\textit{noise term}}. \tag{22}$$

Proof. As in the proof of Proposition 3, $X_n^\top W_i = \sum_k c_{ik}\alpha_{k,n} + \eta_i^\top \varepsilon_n$, and similarly for W_j. Expanding the product and taking $\mathbb{E}[\cdot \mid \mathcal{F}_t]$, the cross terms between $(\alpha_{k,n})_k$ and ε_n vanish (independence, $\mathbb{E}[\varepsilon_n \mid \mathcal{F}_t] = 0$), leaving

$$M_{ij}(t) = \sum_{k,l} c_{ik}c_{jl}\, \mathbb{E}[\alpha_{k,n}\alpha_{l,n} \mid \mathcal{F}_t] + \mathbb{E}\big[(\eta_i^\top \varepsilon_n)(\eta_j^\top \varepsilon_n) \mid \mathcal{F}_t\big].$$

The first sum is the signal term by definition of $p_{kl}(t)$; the second term equals $\sigma_\varepsilon^2\, \eta_i^\top \eta_j$ since $\eta_i, \eta_j \in \mathcal{V}^\perp$ and $\varepsilon_n \sim \mathcal{N}(0, \sigma_\varepsilon^2)$ on $\mathcal{V}^\perp$. □

Pure candidates. When $W_i = V_i$ and $W_j = V_j$, $c_{ik} = \delta_{ik}$, $c_{jl} = \delta_{jl}$, and $\eta_i = \eta_j = 0$: both the signal term collapses to $p_{ij}(t)$ and the noise term vanishes identically, so $M_{ij}(t) = p_{ij}(t)$ exactly. This is just what we had in Proposition 2.

Impure candidates When W_i, W_j are not pure, (22) shows that $M_{ij}(t)$ departs from $p_{ij}(t)$ through two distinct mechanisms, each sufficient on its own to bias the naive measure.

- *Signal leakage.*[CAL: on veut dire "signal leakage" or "narrative leakage"?] The signal term mixes *every* entry of the true co-occurrence matrix $(p_{kl}(t))_{k,l}$, not just $p_{ij}(t)$ itself. In particular, even if narratives i and j never truly co-occur ($p_{ij}(t) = 0$), $M_{ij}(t)$ can be large because W_i and W_j both

carry a nonzero loading on a shared third narrative k ($c_{ik} \neq 0$ and $c_{jk} \neq 0$): the naive approach then reports an artificial synchronization between two narratives that share nothing in the corpus. [CAL: je me demande s'il ne faudrait pas **donner un exemple concret**: these representatives of two distinct narratives (... for narrative A and ... for narrative B) nevertheless carry common concepts like...]

- *Semantic Noise* The noise term, $\sigma_\varepsilon^2 \, \eta_i^\top \eta_j$, is an independent second source of artificial co-variation. Even a candidate pair with perfectly targeted signal ($c_{ik} = \delta_{ik}$, $c_{jl} = \delta_{jl}$, zero leakage onto any other narrative) still registers a spuriously elevated $M_{ij}(t)$ as soon as their residuals η_i, η_j (whatever in each candidate's textual representation is unrelated to any narrative, e.g. shared vocabulary or phrasing used to describe both) happen to be non-orthogonal. Since $\sigma_\varepsilon^2 > 0$, any $\eta_i^\top \eta_j > 0$ mechanically inflates $M_{ij}(t)$ above the true co-occurrence, regardless of whether the narratives themselves ever co-occur in the corpus: two narratives can appear artificially correlated purely because their candidate descriptions happen to be phrased in a similar style.

The naive approach therefore fails on two distinct counts. At the level of prevalence, it conflates a narrative's true presence with leakage from every other narrative sharing representational overlap with it (Proposition 3). At the level of co-occurrence, it conflates genuine joint presence with two sources of spurious correlation: leakage onto narratives, and alignment of narrative-unrelated residuals, induced purely by the geometry of the candidate representations (Proposition 4). Both failure modes vanish exactly, once the candidates are replaced by their pure (orthogonalized) counterparts which is precisely what the procedure of Section 4 constructs.

4. Orthogonalization of Narratives

The previous section demonstrates that *arbitrary representatives of narratives do not have enough good properties to enable to estimate their prevalence in texts*. Nevertheless the modelling angle we exposed in Section 2.2 allows to *establish an orthonormalization procedure allowing to recover a nice candidate collection of pure narratives*. It will be nice in the sense: it faithfully represents the collection of narratives spanned by chosen representatives, and it is pure.

4.1 The Problem: Correlated Narrative Vectors

Let $\{N_i\}_{i=1}^n \subset \mathbb{R}^d$ denote the embedding vectors of n narratives, obtained by passing their textual ~~descriptions~~ representatives through a language model. A fundamental difficulty is that narratives written in the same language, sharing a journalistic style and treating related themes, yield representations that are highly correlated in the embedding space. In our experiments (Section 5), pairwise cosine similarities between raw narrative vectors lie in the range $[0.91, 0.96]$.

A direct consequence is that the time series of cosine similarities $\cos(N_i, X_t)$ (where X_t is the embedding of an article published at time t) evolve almost synchronously across narratives, reflecting global topic volume rather than the relative dominance of specific narratives.

The goal is therefore to construct representative vectors $\{V_i\}_{i=1}^n$ that satisfy:

- **Faithfulness**: each V_i is as close as possible to $N_i^* = N_i / \|N_i\|$;
- **Orthonormality**: $V_i^\top V_j = \delta_{ij}$.

4.2 Monotonous Orthogonalization and Its Limits

A natural candidate is the Gram–Schmidt algorithm. However, Gram–Schmidt is strongly order-dependent: the first vector is unchanged while subsequent vectors undergo increasingly large deformations, introducing an arbitrary bias. Moreover, it does not optimize any global fidelity criterion. Importantly, because the procedure is not incremental, any addition or removal of a

narrative requires recomputing the entire orthogonal basis to guarantee optimality and order-independence.

4.3 Global Procrustes Alignment and Its Limits

[TODO: ~~citer une source pour procruste ? c'est visiblement un résultat très classique~~]→ CAL: oui j'ai mis une ref

A natural alternative is the Orthogonal Procrustes solution (Schönemann 1966), which finds the matrix V that minimizes the global Frobenius norm $\|N^* - V\|_F^2$. While this admits an exact closed-form solution via Singular Value Decomposition (SVD), it maximizes the *sum* of alignments rather than the *minimum*. In the context of highly correlated narratives, Procrustes optimization can sacrifice the representation of minority narratives: it allows individual vectors V_i to be pushed far from N_i^* if doing so slightly improves the overall global alignment. To prevent any single narrative from being discarded or severely distorted, we instead require a max-min formulation.

4.4 Global Optimization on the Stiefel Manifold

To overcome these limitations, we seek the orthonormal family $\{V_i\}_{i=1}^n$ that maximizes the minimum alignment with the original narratives, i.e. we solve the max-min problem:

$$\arg\max_{(V_1,\dots,V_n)} \min_{i=1,\dots,n} N_i^{*\top} V_i \qquad \text{subject to} \quad \begin{cases} \|V_i\| = 1 & \forall\, i, \\ N_i^{*\top} V_i \geq 0 & \forall\, i, \\ V_i^\top V_j = 0 & \forall\, i \neq j. \end{cases} \tag{23}$$

The constraint $N_i^{*\top} V_i \geq 0$ ensures that V_i and N_i^* are on the same side of the hyperplane, i.e. that V_i is a genuine representative of N_i rather than its antipode. The max-min objective guarantees that the algorithm focuses on the worst-case narrative, preventing any single narrative from being sacrificed.

Stacking the row vectors V_i into a matrix $V \in \mathbb{R}^{n\times d}$, the orthonormality constraints are equivalent to $VV^\top = I_n$, i.e. V belongs to the *Stiefel manifold* (Boumal 2023; Wen and Yin 2013):

$$\mathrm{St}(n,d) = \left\{ V \in \mathbb{R}^{n\times d} : VV^\top = I_n \right\}. \tag{24}$$

[TODO: Doit-on garder le détail de la résolution du pb d'optimisation dans le texte ou le mettre dans l'annexe.]→ CAL: oui faisons ça

Because the objective $\min_i(N_i^{*\top} V_i)$ is non-differentiable, we approximate it with the smooth *softmin* parameterized by temperature $\tau > 0$:

$$f(V) = -\tau \log \sum_{i=1}^{n} \exp\left(-\frac{N_i^{*\top} V_i}{\tau} \right). \tag{25}$$

Minimizing f on $\mathrm{St}(n,d)$ is performed via *Riemannian gradient descent*, with three steps per iteration:

1. **Euclidean gradient.**

$$\nabla f(V) = -\operatorname{diag}(w)\, N, \qquad w_i = \frac{\exp(-N_i^{*\top} V_i/\tau)}{\sum_j \exp(-N_j^{*\top} V_j/\tau)}. \tag{26}$$

2. **Riemannian gradient.** Project onto the tangent space of $\mathrm{St}(n,d)$ at V:

$$\operatorname{grad} f(V) = \nabla f(V) - V \operatorname{sym}\left(V^\top \nabla f(V)\right), \qquad \operatorname{sym}(A) = \tfrac{1}{2}(A + A^\top). \tag{27}$$

3. **Retraction.** After the descent step $\tilde{V} = V - \alpha \operatorname{grad} f(V)$, the updated matrix $\tilde{V}$ no longer necessarily satisfies the orthonormality constraint defining the Stiefel manifold (i.e., $\tilde{V}^\top \tilde{V} \neq I$ in general).
   To map $\tilde{V}$ back onto the Stiefel manifold $\mathrm{St}(p, n)$, we apply a retraction based on the QR decomposition. Concretely, we compute the QR factorization $\tilde{V} = QR$, where Q has orthonormal columns and R is upper triangular. The retracted point is then defined as $V_{\text{new}} = Q$.
   As discussed in Absil et al. Absil, Mahony, and Sepulchre 2009, Chapter 4, standard QR-based retractions are used to project iterates back onto the Stiefel manifold after a descent step.

This iterative scheme guarantees that the orthonormality constraints are satisfied at every iteration while converging to a local optimum of (23).

4.5 Interpretation of the Orthogonalized Vectors

We test our procedure with narratives that can be found in Appendix 1. The four panels of Figure 1 and 2 illustrate the behavior of the original narrative embeddings and their orthogonalized counterparts under both the Stiefel-based procedure and the classical Gram-Schmidt algorithm.

Panel (a) displays the cosine similarity matrix between the raw narrative embeddings. A striking feature is the overall high level of correlation across all narrative pairs, reflecting the anisotropy of the embedding space produced by *ModernBERT*. Diagonal blocks are systematically darker, confirming that narratives belonging to the same thematic chapter are more strongly correlated with one another than with narratives from other chapters. For instance, *"AI Disruption of Labor Markets"* and *"Big Tech Race and Regulatory Pushback"* exhibit a cosine similarity of 0.95, whereas the similarity between *"AI Disruption of Labor Markets"* and *"Financial Market Volatility and Fed Signaling"* is lower (with cosine similarity of 0.91). This intra-chapter cohesion validates the semantic consistency of the narrative groupings. The representative naratives take as examples in Section come from this list (see Appendix 1).

Panel (b) demonstrates the success of the Stiefel optimization. The resulting representative matrix is fully orthogonal, with pairwise cosine similarities equal to zero across all narrative pairs. Crucially, this orthogonality holds not only across thematic chapters but also within chapters: *"Extreme Heat and Emergency Policy Response"* is equally orthogonal to *"International Climate Commitments Under Pressure"* and *"US China Rivalry and Indo Pacific"*. The procedure thus isolates the idiosyncratic component of each narrative, stripping away both cross-chapter and intra-chapter shared variance. What remains is the semantic signal that is strictly proper to each narrative, independently of all others.

Panel (c) presents the cross-similarity between raw narratives and their Stiefel-optimized representatives. A clear diagonal structure emerges: each representative is more strongly correlated with its own source narrative than with any other, confirming that the optimization procedure preserves the idiosyncratic semantic content of each narrative while beginning to disentangle inter-narrative correlations.

Panel (d) exposes the limitations of the classical Gram-Schmidt orthogonalization. This procedure is well known to be sensitive to the ordering of input vectors: by construction, the first vector in the sequence is left entirely unchanged, while subsequent vectors are progressively deflated. As a result, the first representative, here that of *"Extreme Heat and Emergency Policy Response"*, retains its original embedding and remains correlated with all other raw narratives. The pathological consequence is particularly striking: the representative of *"Extreme Heat and Emergency Policy Response"* exhibits a higher cosine similarity with the raw narrative *"Middle East Conflict and Humanitarian Crisis"*, for instance, than the representative of *"Middle East Conflict and Humanitarian Crisis"* does with the very narrative it is supposed to represent. This severe distortion renders the Gram-Schmidt basis semantically uninterpretable as a set of narrative representatives, and underscores the superiority of the Stiefel-based approach for this task.

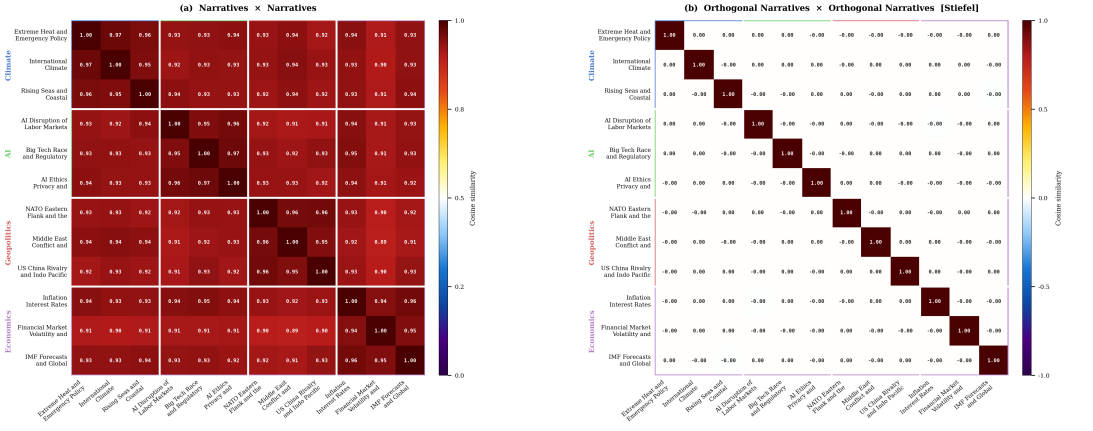


Figure 1. Cosine similarity matrices of narrative embeddings before and after Stiefel orthogonalization. Panel (a) shows the raw *ModernBERT* embeddings, where diagonal blocks are systematically darker, reflecting stronger intra-chapter semantic cohesion. Panel (b) shows the orthogonalized embeddings after Stiefel optimization, yielding zero pairwise cosine similarity across all narrative pairs thereby isolating the idiosyncratic semantic signal of each narrative.

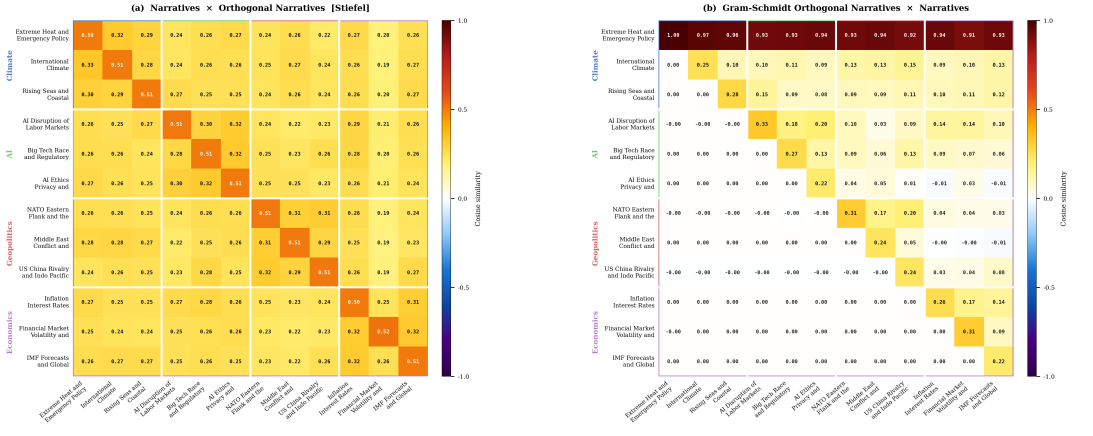


Figure 2. Cross-similarity matrices between raw narrative embeddings and their orthogonalized representatives. Panel (c) shows the Stiefel-optimized representatives, where a clear diagonal structure confirms that each representative retains the idiosyncratic semantic content of its source narrative. Panel (d) shows the Gram-Schmidt representatives, revealing a severe order-dependent distortion: the first narrative is left unchanged and remains correlated with all others, to the point where its representative exhibits higher similarity with unrelated raw narratives than some narratives' own representatives do with themselves.

5. Experiments and Validation on a Dataset

[je propose qu'on laisse tomber ces deux points] [~~CAL: expliquer qu'on utilise cette base car elle va permettre le contrefactuel + mettre en place le framework probabiliste pour les tests~~] [~~CAL: alternatives à modernBERT~~] **[JAMM: ~~première recherche: https://chatgpt.com/share/69e58d30-f2c4-8329-aa4f-5682054e5775~~]**

The objective of this empirical study is to validate the methodology introduced in the previous sections on a labelled dataset from two complementary perspectives. It is not obvious to find a dataset of text that has labels compatibles with narratives; we found the *News Category Dataset* (described in

detail in Section 5.1). What we need is a corpus of texts classified in theme corresponding to potential narratives so that we can check: our orthogonalization procedure produce time series of prevalences that "make sense" (that is a qualitative procedure conducted in Section 5.3), by choosing narratives that are meant to be disjoint with respect to the labelled themes, we can check our orthonormalization procedure respect this *disjunction* (typically orthogonalized narratives in the theme "Entertainment" does not align with orthogonalized narratives in the theme "Politics").

More precisely we will compare those two approaches:

- We use the embeddings that directly encodes the text representations of the studied collection of narratives. We denote them as $(N_i)_i$ and we will sometimes refer to them as "raw narratives ".
- We use the vectors obtained after applying the procedure described in 4.4. We denote them as $(V_i)_i$ and we will sometimes refer to them as "pure narratives "or "orthogonalized narratives ".

We want to check that:

- That orthogonalized narratives effectively remove the biases highlighted in 2;
- The orthogonalized representatives preserve the meaning;
- Tracking a document's similarity to an orthogonalized narrative is a more meaningful way to monitor the exposure to that specific narrative (compared to using the raw narratives)

~~To this end, we rely on the *News Category Dataset*, whose manually assigned categories provide a natural counterfactual framework.~~

5.1 Dataset and Experimental Setup

Dataset. We use the *News Category Dataset* (Misra 2022), which contains approximately 210 000 headlines published by *HuffPost* between 2012 and 2022. This dataset has been for instance used by (Authors of arXiv:2410.10756 2024) for evaluating few-shot selection strategies in LLM-based text augmentation, (Authors of arXiv:2406.10773 2024) to fine-tunes RoBERTa classifiers.

We perform descriptive statistics on this dataset to keep its main categories over the 42 available; it left us with 4 main categories: ~~Among all available categories, we retain four sufficiently represented themes (that we will use as "Chapters" in the terminology of the previous section):~~ *Politics* (32 739 articles), *Entertainment* (16 058), *Travel* (9 887), and *Parenting* (8 677 articles). These four chapters naturally match the narrative groups introduced below and provide labeled subsets that can be exploited for validation. Since the archive becomes sparse after May 2018, we restrict the analysis to the 2012–2018 period, resulting in approximately 200 000 articles.

As shown in Table 2: Entertainment and Politics are the ones stationary enough to test our temporal conditioning, and we keep Parenting and Travel for comparison.

[TODO: ~~Ici, il y a sans doute un commentaire à faire sur les catégories (les seules ok à mon avis c'est entertainment et politics) – ce qui n'empêche pas d'utiliser les autres si on moyenne sur toute la période~~][CAL: oui j'a fait ça]

Temporal resampling. The raw dataset is highly unbalanced over time, particularly for the Politics category during the 2016 presidential campaign. To avoid interpreting variations in publication volume as changes in narrative prevalence, we perform a temporal resampling that enforces a constant weekly number of articles while preserving the relative proportions across categories.Table 2 reports the resulting distribution.

Embeddings. Both news headlines and narrative descriptions are embedded in the same 768-dimensional semantic space using *ModernBERT* (Warner, Mishra, Stolfo, et al. 2024). Narrative descriptions were generated with the assistance of a large language model to ensure a homogeneous writing style.

Table 2. Number of articles per category and year after resampling.

Year	Entertainment	Parenting	Politics	Travel
2012	792	4,087	0	4,221
2013	1,154	4,710	0	3,786
2014	2,445	1,403	3,684	2,079
2015	3,614	0	5,351	705
2016	2,478	0	6,885	311
2017	2,194	0	7,273	178
2018	1,375	0	2,450	60

5.2 Narratives

For each of the four chapters, we manually define three representative narratives intended to capture the main social, political, and cultural themes that characterize the corresponding articles over the 2012–2018 period. The complete list of the twelve narratives is provided in Appendix 2. A selection is shown below.

Table 3. Narratives used in the analysis (selection).

Chapter	Narrative	Description
Politics	Trump as Disruptor and Institutional Stress	Donald Trump rises to power by channeling distrust toward political elites, media institutions, and bureaucratic authority, presenting himself as a disruptive outsider challenging democratic norms.
Entertainment	Entertainment as Political Resistance	Film, television, comedy, and celebrity culture become arenas of political contestation blending entertainment with explicit political messaging.
Parenting	The Overwhelmed Modern Parent	Parenthood is portrayed as emotionally intense and structurally unsupported, with parents struggling to balance careers, childcare costs, and social expectations.
Travel	Experiential and Transformational Travel	Travel is framed as a path to personal growth and cultural immersion through encounters with landscapes, heritage sites, and local communities.

The $(N_i)_i$ raw narrative embedding are then orthogonalized using the Stiefel optimization procedure described in Section 4, producing the corresponding *collection of pure narratives* $(V_i)_i$.

5.3 First qualitative results

We first perform a qualitative comparison between the prevalence estimators obtained using the raw narrative vectors and their orthogonalized representatives.

To facilitate visual interpretation that can be done on Figure 3, we restrict this analysis to the Politics and Entertainment chapters over the 2014–2018 period, where both categories contain a sufficiently balanced number of observations. The displayed curves correspond to an exponential moving average of the estimator $\hat{p}_i(t)$ introduced in Section 2.

The contrast between the two approaches is immediate. Under the raw narrative vectors, the prevalence series evolve almost perfectly synchronously, illustrating the leakage phenomenon anticipated in Section 2: when the prevalence of an unrelated but semantically connected narrative rises, the naive prevalence indicator computed on raw narratives increases mechanically along with



Figure 3. Time series of EMA weekly means of cosine similarities for raw and orthogonal narratives.

it. Pure narratives, by contrast, produce clearly differentiated temporal trajectories, with each of them now exhibiting its own distinctive dynamics.

The clearest visual illustration of this contrast occurs between January 2017 and January 2018, when a pronounced bump appears in the raw prevalence series of virtually every narrative. Under the purified representation, this bump is observed almost exclusively in the Welfare State Under Pressure narrative, the only one for which it plausibly reflects genuine content rather than a shift shared indiscriminately across the corpus. *This single episode captures the practical interest of orthogonalization: without it, an apparent surge in any given narrative could just as easily be an artifact of another, unrelated narrative's rise.* This pattern is consistent with the historical record, as the early months of the Trump administration were marked by sustained legislative pressure on the social safety, most notably through repeated attempts to repeal the Affordable Care Act (Obamacare).

The observed behavior is also consistent with the historical context used to design the narratives. The political narratives were deliberately chosen to capture the rise of Donald Trump's presidential campaign and its associated political themes. Accordingly, the prevalence of the narratives Trump as Disruptor and Institutional Stress and Culture War and Identity Conflict increases steadily throughout the campaign and reaches its highest levels around the 2016 election. This evolution is qualitatively consistent with the growing importance of these themes in the political debate.

Conversely, the Entertainment narratives were intentionally designed to represent long-term structural trends rather than isolated historical events. Their estimated prevalence therefore remains comparatively stable over time, exhibiting only moderate fluctuations. The resulting time series are thus coherent with the expected temporal behavior of the corresponding narratives.

5.4 From Qualitative to Quantitative Virtues of Stiefel Orthonormalizaton

[JAMM: ~~TO DO: changer le titre~~][CAL: c'est fait]

We next evaluate the proposed methodology through three complementary quantitative experiments.

5.4.1 Reduction of artificially high co-variation

We first observe qualitatively that the prevalence curves within the political and Entertainment subsets of narratives evolve almost synchronously. We now turn to quantifying this phenomenon across the full set of narratives under study.

We use the metric introduced in Section 2: [CAL: j'ai enlevé les n (on voit bien qu'il y avait un problème)]

$$M_{ij}(t) := \mathbb{E}\left[(X^\top W_i)(X^\top W_j) \mid \mathcal{F}_t\right].$$

As discussed there, for a collection of raw, non-orthogonalized narratives, two distinct effects artificially inflate this covariation: signal leakage and the correlation induced by the noise term. We therefore assess whether orthogonalization produces a sizable reduction in covariation. Time is discretized weekly: for each pair (i, j) we compute $M_{ij}(t)$ and average it over the entire study period.

Table 4. Descriptive statistics of pairwise narrative covariation $M_{ij}(t)$ ($\times 10^{-7}$), before (N) and after (V) orthogonalization.

	N (raw)	V (purified)
mean	74.018	6.486
std	0.646	0.737
median	73.985	6.421
min	72.775	5.209
max	75.546	8.343

The empirical results strongly support this prediction: as reported in Table 4, pairwise covariation scores computed from the raw narrative vectors are, on average, approximately one order of magnitude larger than those obtained after orthogonalization.

5.4.2 Semantic specificity

The second validation exercise aims to verify that purified representatives preserve the semantic meaning of the original narratives.

The labeled structure of the News Category Dataset naturally provides a counterfactual framework: since every narrative belongs to one of four chapters, we compare its average cosine similarity within its own chapter to its average similarity outside that chapter.

For the orthogonalized representatives, we define

$$\Delta^{i,k}(V_{i,k}) = \mathbb{E}\left[(X^i)^\top V_{i,k}\right] - \mathbb{E}\left[(X^{\neg i})^\top V_{i,k}\right],$$

where X^i denotes articles belonging to chapter i, while $X^{\neg i}$ denotes all remaining articles.

The analogous quantity is computed for the original narrative vectors,

$$\Delta^{i,k}(N_{i,k}) = \mathbb{E}\left[(X^i)^\top N_{i,k}\right] - \mathbb{E}\left[(X^{\neg i})^\top N_{i,k}\right].$$

A meaningful narrative should be more strongly represented within its own chapter than outside it, implying

$$\Delta^{i,k}(N_{i,k}) > 0, \qquad \Delta^{i,k}(V_{i,k}) > 0.$$

More importantly, if orthogonalization successfully removes semantic information shared with other chapters while preserving each narrative's intended meaning, we expect

$$\Delta^{i,k}(V_{i,k}) > \Delta^{i,k}(N_{i,k}).$$

Table 5. Sensitivity analysis: 1000 bootstrap draws, 100% of articles, 95% CI for $\Delta^i(V_i)$, $\Delta^i(N_i)$, and $\Delta_{\text{gain}} = \Delta^i(V_i) - \Delta^i(N_i)$. Having all $\Delta^i(V_i) \geq 0$ is compatible with our theory, whereas the only occurrences of unexpected Δ_{gain}, i.e. negative values, are in red.

Chapter	Narrative	$\Delta^i(N_i)$	$\Delta^i(V_i)$	Δ_{gain}
Politics	Trump as disruptor and institutional stress	0.0019 [0.0015, 0.0023]	0.0377 [0.0374, 0.0381]	0.0358 [0.0353, 0.0363]
Politics	Culture war and identity conflict	−0.0002 [−0.0006, 0.0003]	0.0312 [0.0309, 0.0316]	0.0314 [0.0309, 0.0319]
Politics	Welfare state under press	−0.0053 [−0.0058, −0.0048]	0.0084 [0.0078, 0.0090]	0.0137 [0.0134, 0.0141]
Entertainment	Entertainment as political resistance	−0.0145 [−0.0151, −0.0139]	0.0062 [0.0058, 0.0066]	0.0207 [0.0203, 0.0210]
Entertainment	Franchise nostalgia and serial dominance	−0.0107 [−0.0112, −0.0101]	0.0207 [0.0202, 0.0211]	0.0313 [0.0309, 0.0318]
Entertainment	Celebrity image management in the social media…	−0.0153 [−0.0159, −0.0148]	0.0054 [0.0050, 0.0058]	0.0207 [0.0202, 0.0212]
Parenting	The overwhelmed modern parent	0.0253 [0.0249, 0.0256]	0.0118 [0.0114, 0.0123]	−0.0134 [−0.0139, −0.0129]
Parenting	Intensive and expert guided parenting	0.0287 [0.0283, 0.0291]	0.0302 [0.0297, 0.0308]	0.0015 [0.0010, 0.0020]
Parenting	Raising children in a politicized society	0.0244 [0.0240, 0.0248]	0.0174 [0.0171, 0.0178]	−0.0070 [−0.0073, −0.0066]
Travel	Experiential and transformational travel	0.0251 [0.0246, 0.0257]	0.0298 [0.0293, 0.0304]	0.0047 [0.0042, 0.0052]
Travel	Travel optimization and consumer strategy	0.0279 [0.0274, 0.0285]	0.0415 [0.0410, 0.0420]	0.0135 [0.0130, 0.0141]
Travel	Aesthetic and instagrammable destinations	0.0297 [0.0293, 0.0303]	0.0452 [0.0446, 0.0458]	0.0155 [0.0150, 0.0160]

The results in Table 5 are satisfactory: every chapter exhibits $\Delta^i(V_i) > 0$ [CAL: répéter ce que ça veut dire]. The Entertainment chapter and two narratives of Politics chapter have $\Delta^i(N_i) < 0$. Regarding Δ_{gain}, it is positive for three chapters out of four. [CAL: répéter ce que ça veut dire]

5.4.3 Narrative specificity of detected ruptures

We now quantify the qualitative observation made above: ruptures detected on raw narrative representations tend to be shared across many, otherwise unrelated, narratives, whereas ruptures detected on purified representations are largely specific to a single narrative.

[CAL: je trouve que c'est trop compliqué; par besoin de $Q(M)$ je pense (et il y a encore le problème des n, je mettrai qq chose comme:]

We define a rupture by the rejection by a two-sided Welch t-test of the null hypothesis:[1] "*the prevalence of the narrative M in News before t is statistically not different from the prevalence of the narrative M in News after t*". We define $\mathcal{B}_t$ and $\mathcal{A}_t$ a windows of 90 days respectively before and after t and we test, on the raw narratives N and on the pure narratives V the hypothesis at each date t:

$$H_0(t, M) : \mathbb{E}\left[X^\top M \mid X \in \mathcal{B}_t\right] = \mathbb{E}\left[X^\top M \mid X \in \mathcal{A}_t\right].$$

It means that we obtain one time series $H_0(t, N_i)$ for each raw narrative N_i (upper panel of Figure 4) and one time series $H_0(t, V_i)$ for each corresponding pure narrative V_i (lower panel of Figure 4).

~~For a narrative representation $M \in \{N_i, V_i\}$ and each candidate date t, let $\mathcal{B}_t$ and $\mathcal{A}_t$ denote the sets of articles published, respectively, in the 90-day window preceding and following t. Writing $Q_n(M) = X_n^\top M$ for the projected score of article n, we test, at every candidate date t,~~

$$H_0 : \mathbb{E}\left[Q_n(M) \mid n \in \mathcal{B}_t\right] = \mathbb{E}\left[Q_n(M) \mid n \in \mathcal{A}_t\right],$$

1. Because this test is repeated over hundreds of candidate dates, the resulting p-values are corrected for multiple testing using the Benjamini–Hochberg FDR procedure; consecutive significant dates are then grouped into a single detected rupture

~~using a two-sided Welch t-test, which does not require equal variances across the two windows. Because this test is repeated over hundreds of candidate dates, the resulting p-values are corrected for multiple testing using the Benjamini–Hochberg FDR procedure; consecutive significant dates are then grouped into a single detected rupture.~~ This procedure is applied separately to the raw (N) and purified (V) representations of every narrative in our collection.

Results are reported in Figures 4. Three observations, all compatible with our model, comforting our methodology, stand out:

- First, the number of detected ruptures is systematically higher under N than under V: averaged across narratives, 27.1 ruptures are detected under N, against 16.5 under V, and this ordering holds narrative by narrative without exception.
- Second, pairwise overlap between the sets of rupture dates detected across different narratives is markedly higher under N than under V, as measured by the Jaccard index[2]: the average pairwise Jaccard index is 0.74 under N, compared to only 0.25 under V. Taken together, these two findings confirm that raw representations tend to co-detect the same, corpus-wide ruptures, consistent with the leakage phenomenon described in Section **??**, whereas purified representations isolate ruptures that are substantially more specific to each individual narrative.
- Third, this is directly illustrated by the July 2017 bump discussed in our qualitative analysis: under N, it is detected as a significant rupture for every narrative in the collection, while under V it is detected almost exclusively for the Welfare State Under Pressure narrative, the only one for which it plausibly reflects genuine, narrative-specific content (as the period corresponds to Trump's first semester as President with notably repeated attempt to repeal the Obamacare) rather than a shift shared indiscriminately across the corpus.

6. Conclusion

We have introduced a rigorous algebraic methodology for monitoring the relative prevalence of competing narratives in a time-evolving text corpus.

It allows to estimate the relative prevalence of narratives directly in the space of embedding, that is far cheaper and more reliable than prompting frontier models.

The key insight is that standard cosine similarity between embedding of raw narrative (i.e. arbitrary representative of a narrative using a sentence in natural language) suffers from a structural correlation bias induced by shared semantic content, which we characterize formally through a linear generative model. It allows us to propose a method to orthogonalize arbitrary collection of narratives to turn them into what we call *a collection of pure narratives*. Our remedy is a global orthogonalization on the Stiefel manifold. It eliminates the shared component while preserving each narrative's idiosyncratic direction, yielding an unbiased, non-parametric, and interpretable prevalence estimator. Validation on a large labeled news corpus confirms that the estimator faithfully recovers known historical dynamics, whereas the naive approach fails.

Using a labelled dataset (the *News Category Dataset*), we show qualitatively and quantitatively that our method is sound and allow to draw conclusions that otherwise would have been hidden into artefacts of natural language syntax.

Perspectives. The natural extension of this work is the application of the estimator to large-scale ~~financial news~~ economic corpora, opening the way to a direct quantitative model of narrative competition and its influence on the formation of economic expectations, that is a question at the intersection of computational linguistics and price formation.

2. Jaccard index: $J(A, B) = \dfrac{|A \cap B|}{|A \cup B|}$.

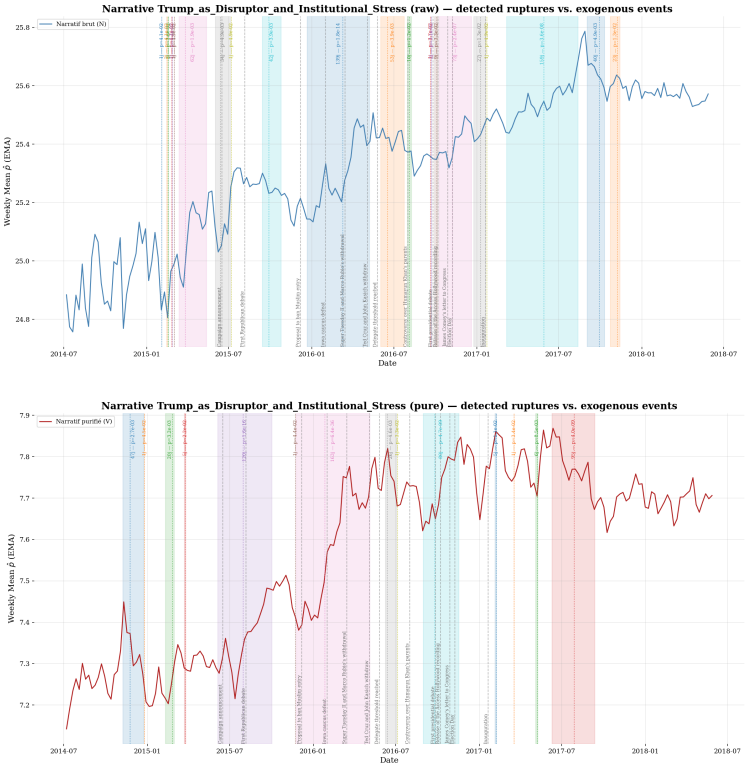


Figure 4. Time series of EMA weekly means of cosine similarities for raw and orthogonal Trump narrative, with detected ruptures as well as a set of exogenous events related to Trump's campaign.

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Competing Interests. None.

Data Availability Statement. The News Category Dataset is publicly available (Misra 2022). Replication code can be found at **[TODO: insert repository URL]**.

Ethical Standards. The research meets all ethical guidelines, including adherence to the legal requirements of the study country.

Author Contributions. The initial work on orthogonalization in the space of embedding of narrative started as a student project of 3rd year of Ecole Polytechnique by Juliette Anglade and Mario Massy under the direction of Pr. Charles-Albert Lehalle. It later evolved in a joined research effort, reported into this paper. The contributions of the three authors is thus equal.

[TODO: Est-ce que c'est léger en termes de références ?] [CAL: je propose de poser cette question à Perplexity une fois qu'on aura la version finale... mais perso je trouve que c'est bien]

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Appendix 1. Complete List of Narratives as an Illustration

Table 6. All narratives used in the analysis.

Chapter	Narrative	Description
Climate	Extreme Heat and Emergency Policy Response	Record-breaking temperatures across southern Europe this summer triggered unprecedented heatwaves, forcing governments to activate emergency climate plans and accelerate the transition toward renewable energy sources in order to meaningfully reduce carbon emissions and limit further global warming.
Climate	International Climate Commitments Under Pressure	As climate-related disasters multiply worldwide, the United Nations urgently calls for stronger adherence to the Paris Agreement, emphasizing that industrialized nations must deliver substantial and verifiable reductions in greenhouse gas emissions before irreversible climate tipping points are reached.
Climate	Rising Seas and Coastal Vulnerability	The accelerating melting of Arctic glaciers, driven by sustained global warming, is causing a gradual but measurable rise in sea levels that threatens tens of millions of people living in low-lying coastal zones, according to a major scientific report released this week.
Artificial Intelligence	AI Disruption of Labor Markets	The rapid rise of generative artificial intelligence models is profoundly reshaping global labor markets, prompting technology companies to invest billions of dollars in autonomous systems capable of gradually replacing skilled workers across a wide range of industries and professional sectors.
Artificial Intelligence	Big Tech Race and Regulatory Pushback	Silicon Valley's leading technology firms are intensifying their competition in artificial intelligence by unveiling increasingly powerful language models, while European regulators work to constrain these developments through strict legislative frameworks designed to protect citizens' fundamental rights and digital freedoms.
Artificial Intelligence	AI Ethics, Privacy, and Public Sector Governance	Recent breakthroughs in artificial intelligence are raising critical ethical concerns around personal data privacy and cybersecurity, prompting several national governments to establish dedicated regulatory frameworks governing the deployment of these technologies within public institutions and essential services.
Geopolitics	NATO Eastern Flank and the Russia–Ukraine War	The ongoing military conflict between Russia and Ukraine continues to destabilize the European geopolitical order, compelling NATO allies to reinforce their eastern flank with additional troops and to significantly increase national defense budgets in response to sustained and escalating Russian aggression.
Geopolitics	Middle East Conflict and Humanitarian Crisis	The armed conflict in the Middle East has triggered a humanitarian catastrophe of alarming proportions, prompting urgent international calls for an immediate ceasefire and the rapid deployment of emergency aid to civilian populations trapped in active combat zones across the region.

Continued on next page

Table 6 – *Continued from previous page*

Chapter	Narrative	Description
Geopolitics	US–China Rivalry and Indo-Pacific Tensions	Growing geopolitical rivalry between Washington and Beijing over the status of Taiwan is seriously threatening stability across the Indo-Pacific region, with both superpowers escalating military demonstrations and economic sanctions in what increasingly resembles a renewed cold war confrontation.
Economics	Inflation, Interest Rates, and Household Strain	Persistent inflation across the world's major economies is forcing central banks to maintain elevated interest rates, placing significant pressure on economic growth and disproportionately burdening lower-income households struggling to absorb the rising cost of everyday goods and essential services.
Economics	Financial Market Volatility and Fed Signaling	Global financial markets experienced sharp volatility this week as investors reacted to the latest monetary policy signals from the United States Federal Reserve, triggering significant fluctuations across major stock indices and a notable depreciation of several emerging market currencies.
Economics	IMF Forecasts and Global Growth Slowdown	Global economic growth is decelerating at a concerning pace according to the latest International Monetary Fund projections, which identify trade tensions, rising public debt levels, and slowing emerging market performance as the primary drivers behind deteriorating worldwide economic prospects.

Appendix 2. Complete List of Narratives for the Analysis

Table 7. All narratives used in the analysis.

Chapter	Narrative	Description
Politics	Trump as Disruptor and Institutional Stress	Donald Trump rises to power by channeling distrust toward political elites, media institutions, and bureaucratic authority, presenting himself as a disruptive outsider challenging democratic norms and reframing institutional resistance as partisan sabotage.
Politics	Culture War and Identity Conflict	American politics becomes increasingly structured around culture wars in which abortion, LGBTQ rights, immigration, race, religion, and gun control become symbolic battlegrounds mobilizing voters through moralized political conflict.
Politics	Welfare State Under Press	Political debates around health care reform, social assistance, and regulatory protections reflect a broader struggle over the future of the welfare state amid economic insecurity and ideological polarization.
Entertainment	Entertainment as Political Resistance	Film, television, comedy, and celebrity culture become arenas of political contestation where cultural production increasingly blends entertainment with explicit political messaging.
Entertainment	Franchise Nostalgia and Serial Dominance	The entertainment industry relies heavily on sequels, reboots, and cinematic universes, capitalizing on nostalgia and established IP in an increasingly competitive streaming environment.
Entertainment	Celebrity Image Management in the Social Media Age	Celebrities navigate a hyper-visible media ecosystem where personal relationships, identity declarations, and scandals shape public image and reputation management.
Parenting	The Overwhelmed Modern Parent	Parenthood is portrayed as emotionally intense and structurally unsupported, with parents struggling to balance careers, childcare costs, and social expectations.
Parenting	Intensive and Expert-Guided Parenting	Parenting becomes increasingly expert-driven, with reliance on psychological advice, scientific studies, and educational technologies to optimize children's development.
Parenting	Raising Children in a Politicized Society	Parents confront a social environment shaped by political polarization, gun violence debates, and shifting gender norms that influence children's values and civic identity.
Travel	Experiential and Transformational Travel	Travel is framed as a path to personal growth and cultural immersion through encounters with landscapes, heritage sites, and local communities.
Travel	Travel Optimization and Consumer Strategy	Travel is increasingly treated as a strategic consumer activity involving price optimization, loyalty programs, and efficiency strategies.
Travel	Aesthetic and Instagrammable Destinations	Destinations gain visibility through their visual appeal and shareability on social media platforms, emphasizing photogenic landscapes and unique experiences.